varies in size and shape. They are broadly divided into two main groups.

a. **Yeasts:** The yeasts are unicellular organisms which reproduce by asexual process known as **budding** or by **fission**. The cell develops a protuberance that enlarges and separates from the parental cell. The yeasts produce chains of elongated cells known as **Pseudohyphae**. Some yeasts are reproduced by sexual process Example: *Cryptococcus neoformans*. Germ tube is special morphology found in *Candida albicans*. Some are commensal without any medical significance.

b. **Molds:** The molds grow by apical extension, forming an interwoven mass called as **Mycelium,** branching filaments known as **hyphae**. Hyphae that grow on the surface are called **vegetative hyphae**. They are responsible for the absorption of nutrients. The hyphae that project above the surface are called **aerial hyphae** and they produce specialized reproductive structures called as **conidia**.

HOTS

Can you cultivate the molds at home?

Depending on cell morphology fungi are divided into four types, they are **Yeasts:** These are unicellular organisms that divide by budding (Figure 9.1a & b). Example: *Cryptococcus neoformans* (Pathogenic), *Saccharomyces cerevisiae* (Non pathogenic). **Yeast – like fungi:** These fungi reproduce by budding but fails to separate and hence elongation takes place forming pseudohyphae. Example: *Candida* species (Pathogenic). **Molds:** These fungi produce spores which germinate to form vegetative hyphae (Figure 9.2). Example: Dermatophytes, *Aspergillus, Penicillium, Mucor*. **Dimorphic fungi:** These Fungi exist in both yeast at 37°C and filamentous form at 25°C. This Phenomenon is known as **Fungal dimorphism** (Figure 9.3). Example: *Histoplasma capsulatum*, *Blastomyces dermatitidis*.

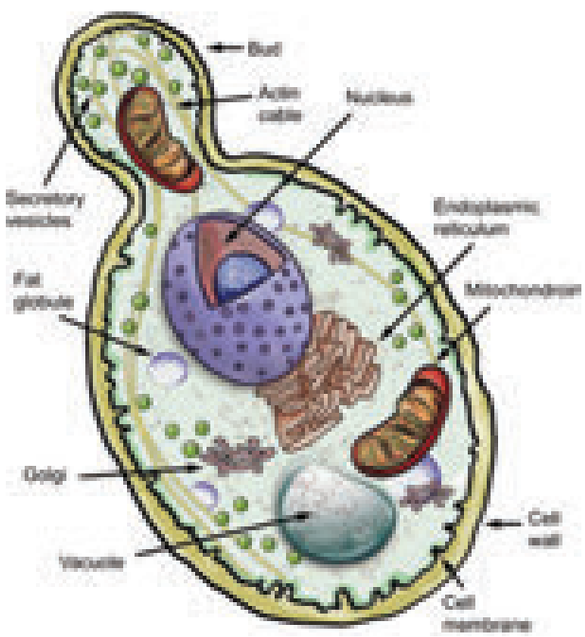
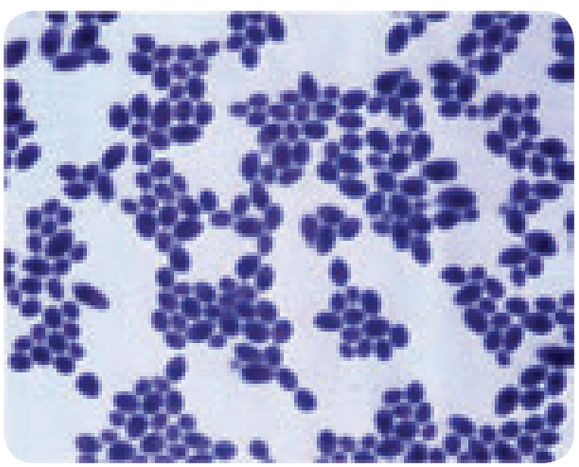


Figure 9.1: (a) Morphology of yeast



(b) Microscopic view of yeast

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